# Experiment 3

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**Aim :** Apply Decision Tree and Random Forest for classification tasks

**Data Source : (Used Car Price Prediction Dataset)**

- Dataset Type: Structured tabular dataset
- File Format: CSV (Comma Separated Values)
- Total Records (Rows): 7,400
- Total Attributes (Columns): 29
- Domain: Used Car Sales / Automobile Marketplace Data
- Dataset Link

**Theory :**
Classification is a supervised machine learning task used to predict categorical outcomes (such as Yes/No, Pass/Fail, Spam/Not Spam). Two popular and powerful algorithms used for classification are **Decision Tree** and **Random Forest**. These algorithms are widely used because they are easy to understand, efficient, and provide good accuracy.

**Decision Tree for Classification :**

A **Decision Tree** is a tree-structured model used to make decisions based on feature values. It splits the dataset into smaller subsets based on certain conditions.

**Structure of Decision Tree**

- **Root Node** – Represents the entire dataset.
- **Internal Nodes** – Represent decision rules based on features.
- **Branches** – Represent outcomes of decisions.
- **Leaf Nodes** – Represent final class labels.

**Working of Decision Tree**

1. Select the best feature using measures like:
    - **Entropy**
    - **Information Gain**
    - **Gini Index**
2. Split the dataset based on the selected feature.
3. Repeat the process recursively.
4. Stop when:
    - All data belong to the same class, or
    - No more features are available.

**Advantages**

- Easy to understand and interpret.
- No need for data normalization.
- Handles both numerical and categorical data.

**Disadvantages**

- Prone to **overfitting**.
- Unstable (small changes in data may change the tree).

**Random Forest for Classification :**

A **Random Forest** is an ensemble learning technique that builds multiple decision trees and combines their predictions to improve accuracy and reduce overfitting.

**Working of Random Forest**

1. Randomly select subsets of the dataset (Bootstrap sampling).
2. Build multiple decision trees.
3. At each split, select random subsets of features.
4. Each tree gives a prediction.
5. Final prediction is based on **majority voting**.

**Advantages**

- Higher accuracy than a single decision tree.
- Reduces overfitting.
- Handles large datasets effectively.

**Disadvantages**

- More computationally expensive.
- Less interpretable compared to a single decision tree.

**Steps to Apply Decision Tree and Random Forest :**

1. Import required libraries (e.g., sklearn).
2. Load dataset.
3. Split the dataset into training and testing sets.
4. Train the model using:
    - DecisionTreeClassifier
    - RandomForestClassifier
5. Predict the test data.
6. Evaluate using accuracy, confusion matrix, precision, recall, F1-score.

**Hyperparameter Tuning :**

**Best Decision Tree Parameters**

- `max_depth = 5`
- `criterion = gini`
- `random_state = 42`

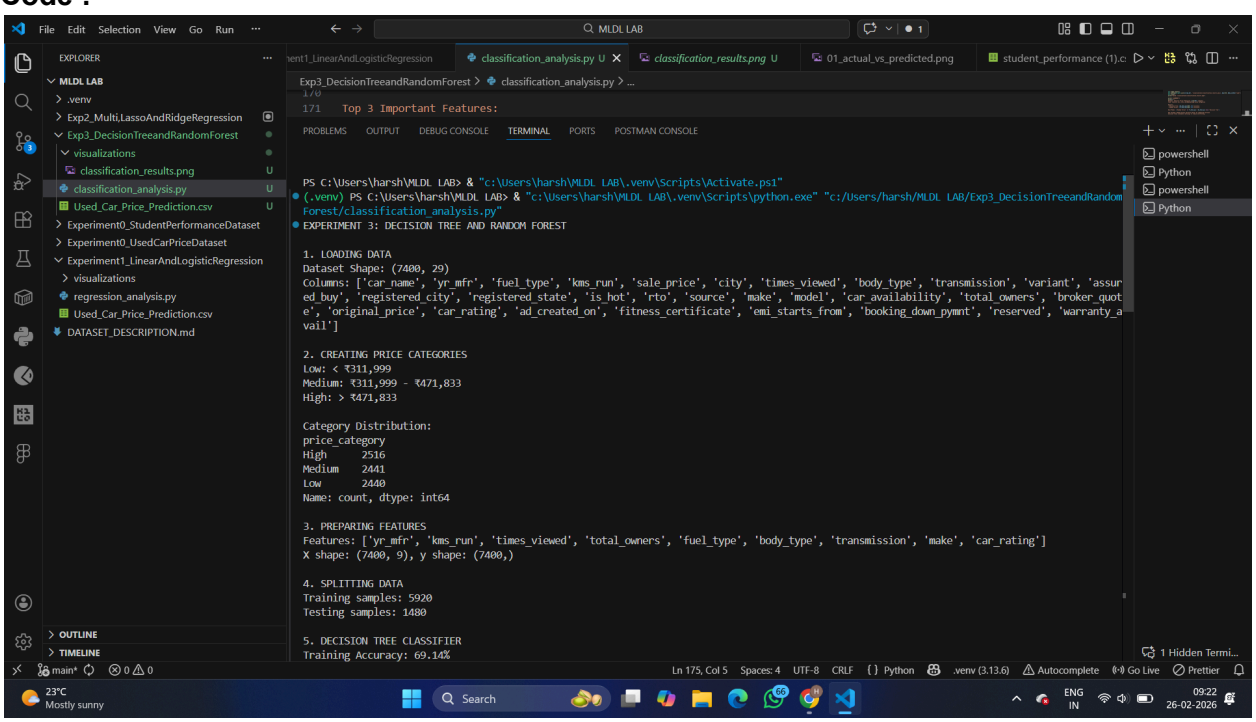
### Best Random Forest Parameters

- `n_estimators = 100`
- `max_depth = 10`
- `max_features = sqrt`
- `bootstrap = True`
- `random_state = 42`

### Tuning Impact :

| Model | Before Tuning | After Tuning | Improvement |
|---|---|---|---|
| Decision Tree | 64.80% | 68.72% | +3.92% |
| Random Forest | 70.07% | 72.09% | +2.02% |

### Code :



```
Top 3 Important Features:

(.venv) PS C:\Users\harsh\MLDL LAB> & "c:\Users\harsh\MLDL LAB\.venv\Scripts\python.exe" "c:/Users/harsh/MLDL LAB/Exp3_DecisionTreeandRandomForest/classification_analysis.py"
5. DECISION TREE CLASSIFIER
Training Accuracy: 69.14%
Testing Accuracy: 68.72%

Classification Report:
              precision    recall  f1-score   support

        High       0.76      0.76      0.76       473
         Low       0.74      0.76      0.75       515
      Medium       0.55      0.54      0.55       492

    accuracy                           0.69      1480
   macro avg       0.69      0.69      0.69      1480
weighted avg       0.69      0.69      0.69      1480


6. RANDOM FOREST CLASSIFIER
Training Accuracy: 81.62%
Testing Accuracy: 72.09%

Classification Report:
              precision    recall  f1-score   support

        High       0.80      0.79      0.80       473
         Low       0.78      0.77      0.77       515
      Medium       0.59      0.61      0.60       492

    accuracy                           0.72      1480
   macro avg       0.72      0.72      0.72      1480
weighted avg       0.72      0.72      0.72      1480
```

```
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(.venv) PS C:\Users\harsh\MLDL LAB> & "c:\Users\harsh\MLDL LAB\.venv\Scripts\python.exe" "c:/Users/harsh/MLDL LAB/Exp3_DecisionTreeandRandomForest/classification_analysis.py"

7. MODEL COMPARISON
Model               Train Acc    Test Acc
Decision Tree       69.14        68.72
Random Forest       81.62        72.09

8. FEATURE IMPORTANCE (Random Forest)
     Feature  Importance
      yr_mfr    0.366279
   body_type    0.169182
     kms_run    0.105687
times_viewed    0.091322
        make    0.084542
   fuel_type    0.084222
  car_rating    0.060758
transmission    0.019620
total_owners    0.018389

9. GENERATING VISUALIZATIONS
Saved: visualizations/classification_results.png

SUMMARY

Dataset: Used Car Price Prediction (7400 samples)
Task: Classify cars into Low/Medium/High price categories

Results:
- Decision Tree: 68.72% accuracy
- Random Forest: 72.09% accuracy

Best Model: Random Forest
```

```
(.venv) PS C:\Users\harsh\MLDL LAB> & "c:\Users\harsh\MLDL LAB\.venv\Scripts\python.exe" "c:/Users/harsh/MLDL LAB/Exp3_DecisionTreeandRandomForest/classification_analysis.py"

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Best Model: Random Forest

Key Insight: Random Forest performs better by combining multiple
decision trees (ensemble learning) to reduce overfitting.

Top 3 Important Features:
1. yr_mfr (0.366)
2. body_type (0.169)
3. kms_run (0.106)

EXPERIMENT COMPLETED!
(.venv) PS C:\Users\harsh\MLDL LAB>
```
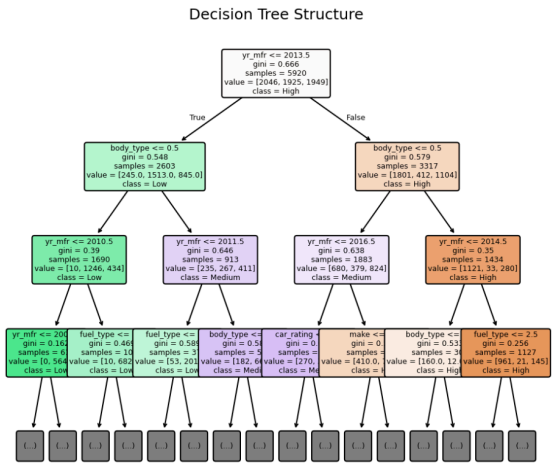
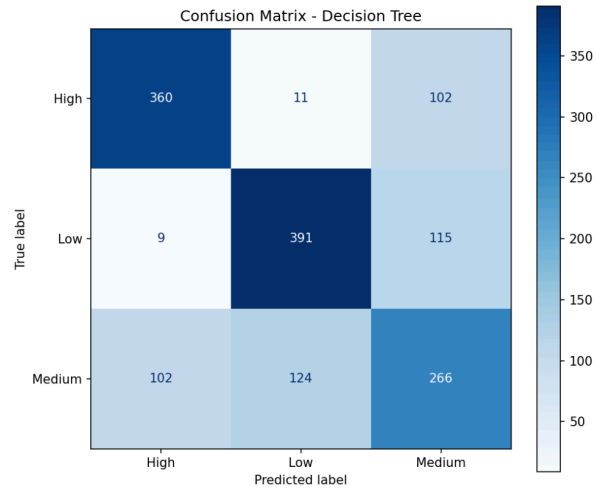
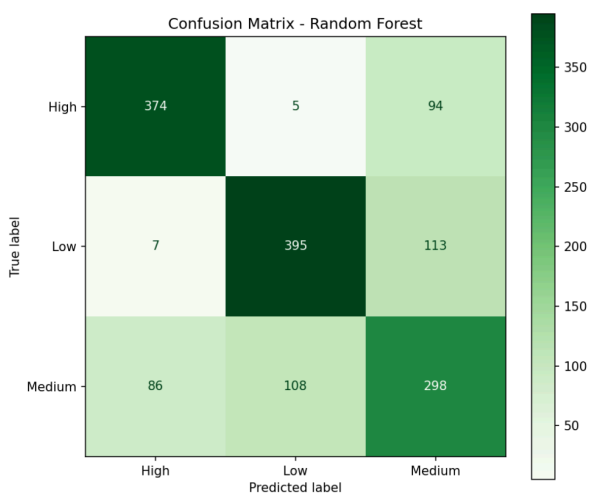
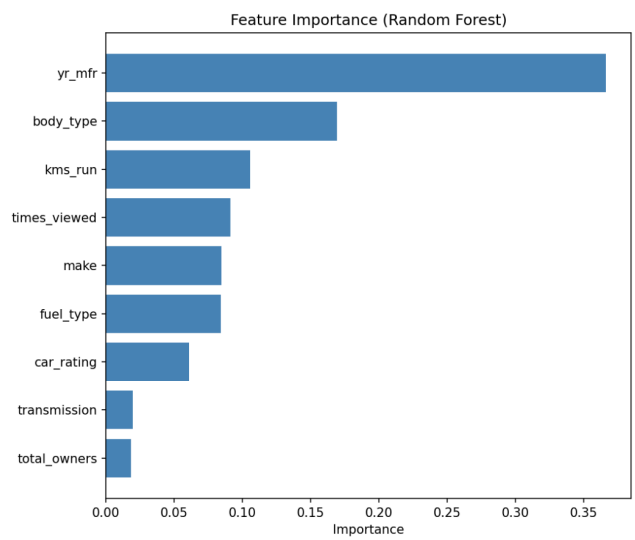

**Output :**




**Conclusion :** Decision Tree and Random Forest are powerful classification algorithms. A Decision Tree is simple and easy to interpret but may suffer from overfitting. Random Forest improves performance by combining multiple trees and using majority voting, making it more accurate and stable. For practical applications, Random Forest is generally preferred when higher accuracy is required, while Decision Tree is useful when model interpretability is important.